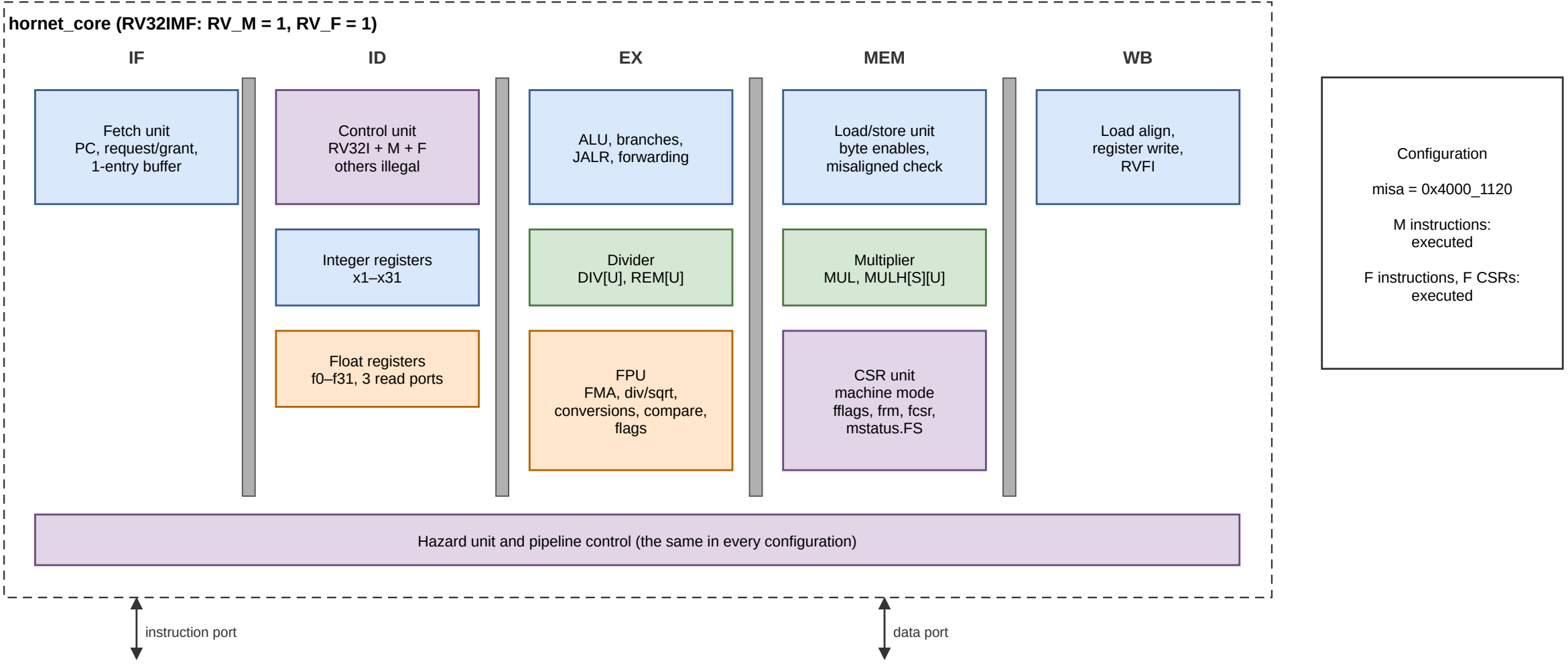
hornet_core (RV32IMF: RV_M = 1, RV_F = 1)
IF
ID
EX
MEM
WB
Fetch unit
PC, request/grant,
1-entry buffer
Control unit
RV32I + M + F
others illegal
Integer registers
x1–x31
Float registers
f0–f31, 3 read ports
ALU, branches,
JALR, forwarding
Divider
DIV[U], REM[U]
FPU
FMA, div/sqrt,
conversions, compare,
flags
Load/store unit
byte enables,
misaligned check
Multiplier
MUL, MULH[S][U]
CSR unit
machine mode
fflags, frm, fcsr,
mstatus.FS
Load align,
register write,
RVFI
Hazard unit and pipeline control (the same in every configuration)
Configuration
misa = 0x4000_1120
M instructions:
executed
F instructions, F CSRs:
executed
instruction port
data port